Buca Eğitim Fakültesi Dergisi,
2026, sayı 67 , ss.571-598

Araştırma Makalesi

The Journal of Buca Faculty of
Education, 2026, issue 67, pp.571-598

Resarch Article

Matematik Eğitiminde Sonsuzluğa Bir Bakış: Bir Betimsel İçerik Analiz Çalışması*

A Glimpse into Infinity in Mathematics Education: A Descriptive Content Analysis Study

Ümranay Kuşçu[1], Serkan Narlı[2]

[1]Sorumlu Yazar, Doktora Öğrencisi, Dokuz Eylül Üniversitesi, umranaykuscu99@gmail.com, (https://orcid.org/0009-0009-0715-1681)
[2]Prof. Dr., Dokuz Eylül Üniversitesi, serkan.narli@deu.edu.tr, (https://orcid.org/0000-0001-8629-8722)

Geliş Tarihi: *18.11.2025* ***Kabul Tarihi:*** *11.03.2026*

ÖZ

Bu araştırmanın amacı, eğitimde sonsuzluk kavramı ile ilgili yapılmış araştırmaların eğilimlerini betimsel içerik analizi yöntemi ile ortaya koymaktır. Bu amaçla, çeşitli veri tabanları taranarak 1956-2024 yılları arasında gerçekleştirilen 224 çalışma analiz edilmiştir. Çalışmalar; yayın yılları, yayın türü, araştırma yöntemi, örneklem özellikleri, veri toplama araçları ve araştırma sonuçları açısından incelenmiştir. Bulgular; en fazla çalışmanın 2006-2010 yılları arasında yapıldığını ve nitel araştırma yöntemlerinin çoğunluğu oluşturduğunu ortaya koymuştur. Örneklem dağılımına bakıldığında, üniversite öğrencilerinin en sık çalışılan grup olduğu görülmüştür. Amaçları doğrultusunda, teorik bilgi sunmayı hedefleyen çalışmaların baskın olduğu, buna karşın sonsuzluk kavramının öğretimine odaklanan araştırmaların sınırlı kaldığı belirlenmiştir. Analiz ayrıca, öğrenenlerin formal eğitim öncesinde ve sonrasında farklı şemalar geliştirdiğini ve önceki öğrenmelerin ileri matematiksel konularda epistemolojik bir engel teşkil edebileceğini göstermiştir. Bu çalışma, erken yaş gruplarına yönelik araştırmalardaki önemli bir boşluğu ortaya koyarak ve deneysel çalışma ihtiyacını vurgulayarak literatüre katkı sağlamaktadır. Sonuç olarak, gelecek araştırmaların, formal tanımlar tanıtılmadan önce sezgisel anlayışı geliştirmek amacıyla sonsuzluk kavramının ilkokul ve ortaokul öğretim programlarına entegre edilmesine odaklanması önerilmektedir.

Anahtar Kelimeler: Betimsel içerik analizi, gerçek sonsuzluk, potansiyel sonsuzluk, sonsuzluk algısı.

ABSTRACT

The purpose of this study is to examine trends in research regarding the concept of infinity in education using a descriptive content analysis method. To this end, 224 studies conducted between 1956 and 2024 were analyzed by scanning various databases. The studies were examined in terms of publication year, type, research method, sample characteristics, data collection tools, and research outcomes. The findings revealed that the highest number of studies was conducted between 2006 and 2010, and qualitative research methods constituted the majority. Regarding sample distribution, university students were the most frequently studied group. In terms of purpose, studies aimed at providing theoretical information were predominant, whereas research focusing on teaching the concept of infinity was limited. The analysis also indicated that learners develop different schemas before and after formal education, and prior learning can act as an epistemological obstacle in advanced mathematical topics. This study contributes to the literature by revealing a significant gap in research regarding early age groups and highlighting the need for experimental studies. Consequently, it is recommended that future research focus on integrating the concept of infinity into primary and middle school curricula to foster intuitive understanding before formal definitions are introduced.

Keywords: Descriptive content analysis, actual ınfinity, potential ınfinity, perception of ınfinity.

INTRODUCTION

The concept of infinity has been one of the concepts that people working in various fields (mathematics, philosophy, religion) throughout history have thought about. Since ancient Greek times, the state of continuity and permanence associated with operations such as counting, ordering, and division has also brought with it the concept of infinity (Allen, 2000). The paradoxes of Zeno play a significant role in the historical development of the concept. For instance, in the famous 'Achilles and the Tortoise' paradox, Zeno asserts that the swift Achilles can never overtake the slow tortoise. This is because, by the time Achilles reaches the point where the tortoise began, the tortoise will have moved a bit further, a process that repeats ad infinitum. Such paradoxes highlight the counter-intuitive nature of infinity (Fischbein et al., 1979). Aristotle also explained the concept of infinity. Aristotle defined infinity as an endless, imperceptible, and incomplete structure (Allen, 2000).

In addition, Aristotle distinguished between two types of infinity: potential infinity and actual infinity (Moreno & Waldegg, 1991). From a philosophical and educational perspective, potential infinity refers to a process that never ends, whereas actual infinity refers to a completed whole (Fischbein, 2001). To illustrate this distinction with a concrete mathematical example, potential infinity can be observed in the counting process (1, 2, 3, …) which grows indefinitely but is never completed (Tall, 1980). In this view, infinity is a 'becoming' process. In contrast, actual infinity is exemplified by the set of all natural numbers ($\mathbb{N} = \{1, 2, 3, \ldots\}$) conceived as a single, static, and completed entity containing all elements simultaneously (Fischbein, 2001). Providing such examples helps to cognitively structure the distinction between these two types of infinity.

The historical development of the concept of infinity followed a trajectory from metaphysical speculation to mathematical formalization. While Aristotle's rejection of actual infinity dominated scientific thought for centuries, the paradoxes noted by Galileo—such as the one-to-one correspondence between natural numbers and their squares—signaled the need for a new mathematical framework. This framework was established by Georg Cantor in the late 19th century. Cantor's Set Theory provided the mathematical foundations of infinity, moving beyond vague philosophical descriptions (Tall, 2001).

Cantor defined infinity not as a never-ending process, but through the concept of 'cardinality' based on one-to-one correspondences (bijections). For instance, although the set of even numbers ($\mathbb{E} = \{2, 4, 6, \ldots\}$) appears to be a proper subset of the natural numbers ($\mathbb{N} = \{1, 2, 3, \ldots\}$), Cantor demonstrated that they have the same cardinality because a bijection $f(n) = 2n$ can be established between them. He termed this smallest level of actual infinity 'aleph-null' ($\aleph_0$). Furthermore, Cantor proved that not all infinities are equal; for example, the set of real numbers ($\mathbb{R}$) is uncountable and has a greater cardinality than $\mathbb{N}$ (Fischbein, 2001). This mathematical treatment allows for a structured understanding of infinity, distinguishing it from earlier philosophical interpretations.

Throughout history, the concept of infinity has been acknowledged as epistemologically challenging due to its abstract nature (Tall, 1981). Consequently, it is a subject that requires careful pedagogical attention from teachers to help students overcome intuitive obstacles and grasp the formal definitions (Fischbein, 2001; Monaghan, 2001). Although the concept of infinity is not treated as a separate learning domain in the K-12 mathematics curriculum implemented in Türkiye (MEB, 2018), it is implicitly embedded in the foundations of many topics. While not explicitly named in early grades, the concept forms the basis of subjects such as the number line, geometric rays, and rational numbers. In higher education levels, it becomes more explicit in topics like limits, series, and integrals. Similarly, in international frameworks such as the NCTM

standards, infinity is often addressed within the context of 'processes' rather than as an isolated topic (NCTM, 2000). This implicit structure necessitates a careful analysis of how the concept is integrated into teaching practices. At this point, the concept of infinity has been encountered by students, but is generally perceived at an intuitive level. Studies have found significant differences between students' intuitive perceptions and formal definitions of infinity, resulting in various misconceptions (Khasawneh, 2020; Singer & Voica, 2003; Tsamir, 2001). Studies have shown that the concept of infinity is primarily perceived as "a huge number," "an unfinished process," and "something without an end," making it challenging to grasp infinity in a mathematical sense (Falk, 2010; Mamolo & Zazkis, 2008).

According to Tall (2001), mathematical thinking requires transitioning between conceptual knowledge, formal knowledge, and symbolic knowledge; however, this transition can be even more challenging, especially with abstract topics. Infinity is one such complex topic. Individuals often encounter the concept through symbols, but struggle to grasp the meaning these symbols represent (Dubinsky et al., 2005). Numerous studies conducted to date have shown that the difficulties individuals experience in grasping the concept of infinity arise not only from the nature of the idea but also from the organization of teaching processes (Maria et al., 2009; Sbaraglı, 2006).

Recent studies conducted have further explored these challenges by focusing on modern instructional environments and curriculum analysis. For instance, the use of technology-enhanced settings, such as WebQuest-based learning, has been shown to provide new opportunities for pre-service teachers to distinguish between countability and the infinite nature of sets (Göktepe Yıldız & Göktepe Körpeoğlu, 2018).These digital environments allow learners to engage with abstract mathematical objects in a more structured way. Furthermore, recent research highlights that epistemological obstacles regarding infinity, uncertainty, and undefined operations continue to persist among students in both education and science faculties, suggesting a need for deeper conceptual integration in higher education (Oflaz & Polat, 2022).

From a curricular perspective, the implicit presence of infinity in primary and middle school mathematics remains a significant area of discussion. Recent investigations into pre-service teachers' views suggest that while infinity is not explicitly named as a standalone topic, it is fundamentally embedded in objectives related to number sets, patterns, and irrational numbers (Ergene, 2021). These findings confirm that the cognitive transition from intuitive structures to formal definitions, as emphasized by earlier researchers (Narlı & Narlı, 2012), remains a central theme in mathematics education. By synthesizing these current trends, it becomes clear that bridging the gap between students' everyday intuitions and formal mathematical infinity requires a multi-dimensional approach involving technology, curriculum revision, and targeted pedagogical interventions.

Individuals' prior experiences and knowledge often conflict with the information taught in formal education. Individuals possess specific intuitive structures regarding the concept of infinity before beginning formal education; when these structures cannot be integrated with the information acquired through formal education, they turn into misconceptions (Özmantar, 2010). According to Singer and Voica (2003), these intuitive representations must be clearly defined during teaching and integrated into the teaching process. Instead of evaluating students' existing knowledge as incorrect, teachers should create more opportunities for students' cognitive development and improve their thinking skills (Bozkuş, 2014).

Infinity is a highly controversial and abstract concept in mathematics. It acts as an 'epistemological obstacle' that creates cognitive conflicts, especially regarding the difference between potential and actual infinity (Tall, 1980). Issues such as students' misconceptions and attitudes have been a major focus in mathematics education (Fischbein, 2001). However, existing studies are often disconnected and focus mostly on small groups or specific time periods. This

fragmentation hinders a comprehensive understanding of how theoretical debates and pedagogical strategies have evolved to address these universal difficulties.

Therefore, the significance of this study extends beyond merely addressing a deficit in the national literature. It aims to contribute to the field by providing a systematic framework to guide future research. Specifically, there is very limited research on how infinity is introduced to younger students (primary and middle school). This lack of research is a critical issue because early misconceptions can prevent students from understanding advanced topics later (Monaghan, 2001). By synthesizing these trends, this study aims to guide researchers to move from simply 'finding errors' to 'developing practical teaching models' that can help students overcome these cognitive obstacles.

The problem addressed in this research is defined as follows: “What are the trends in studies conducted on the concept of infinity in education?” To answer this main problem, the sub-problems to be addressed are as follows:

1. How is the distribution of the studies on the concept of infinity according to the years of publication?

2. How is the distribution of studies on the concept of infinity according to publication types?

3. How is the distribution of the studies on the concept of infinity according to the research method used?

4. How is the distribution of the studies on the concept of infinity according to the type of sample used?

5. How is the distribution of the studies on the concept of infinity according to the sample size used?

6. How is the distribution of the studies on the concept of infinity according to the data collection tools used?

7. How are the examined studies distributed according to their indexing status and publication journals?

8. How are the primary research aims of the examined studies distributed in terms of their thematic focus?

9. How are the key findings of the examined studies distributed across different thematic categories?

METHODOLOGY

In this study, the descriptive content analysis design, one of the non-interactive designs of the qualitative research approach, was preferred in relation to the process and topics covered. The primary objective of studies employing descriptive content analysis is to identify general research topics, thereby guiding future scientific research (Çalık & Sözbilir, 2014). Descriptive content analysis research is a systematic study that provides a meaningful definition and evaluation of the trends and results of published studies related to the topic under investigation (Jayarajah et al., 2014; Suri & Clarke, 2009). It can be distinguished from content analysis, which is used to analyze data from any study. Content analysis is the process of grouping material into codes and categories to determine the frequency of material use (Gökçe, 2006). Descriptive content analysis is a method that aims to identify trends and findings in studies on common topics (Çalık et al., 2008). Furthermore, since the descriptive content analysis method covers both qualitative and

quantitative studies, it differs from meta-analysis and meta-synthesis methods (Lin et al., 2014; Ültay et al., 2021). The results obtained in this process are intended to guide future research on the concept of infinity (Lune & Berg, 2017; Yıldırım & Şimşek, 2006). In line with the study's purpose, literature studies related to infinity were analyzed, and their orientations were systematically evaluated.

2.1. Data Collection

To access studies related to the concept of infinity in education within the scope of the research, a keyword search was conducted on the internet using the Turkish and English terms 'sonsuzluk', 'sonsuzluk algısı', 'sonsuz kümeler', 'infinity', 'sense of infinity', and 'infinite sets' in the title and abstract sections. The identified keywords were searched in the databases specified in the PRISMA Flow Diagram.

2.1.1. Inclusion and Exclusion

The scope of this study was defined to present a holistic perspective on the concept of infinity in mathematics education. In this context, the analysis encompasses both national and international literature. Accordingly, the inclusion criteria determined for the selection of studies are listed below:

(1) In their 1979 study, Fischbein et al. implicitly suggest that Piaget and Inhelder's 1956 study can be considered the beginning of studies on individuals' understanding of infinity (Monaghan, 2001). Based on this, 1956 was determined as the starting point for the studies included in the research. The research encompasses studies on the concept of infinity in education from 1956 to 2024.

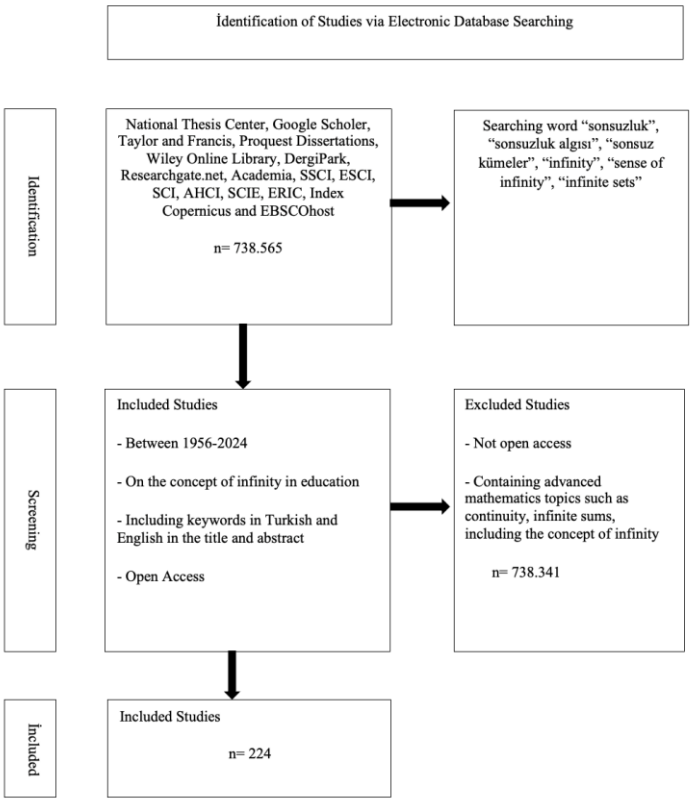
(2) Studies on advanced mathematical topics such as continuity and infinite sums, which involve the concept of infinity, were excluded from the study.

(3) Keyword searches were conducted in Turkish and English on the titles and abstracts of the studies. The research is limited to studies that are accessible and available. To access the studies, keyword searches were conducted on the internet, and theses related to the subject, papers published at conferences, open-access books, and articles accessed through university education faculty and social sciences journals were utilized. It is assumed that all theses, articles, books, and papers published between 1956 and 2024 related to infinity have been accessed.

The PRISMA Flow Diagram used in the search process is shown in Figure 1.

Figure 1

PRISMA Flow Chart



2.1.2. Data Collection Tool

In the study, a 'Study Evaluation Form' developed by the researcher was used to collect data systematically. This form was designed specifically to address the sub-problems of the research and ensures that each study is analyzed within a standard framework. The form consists of specific variables categorized into three main dimensions:

Bibliometric Data: Publication year, publication source (journal/index), and language.

Methodological Characteristics: Research method, research design, and sample group.

Thematic Content: Research aims and key outcomes/findings.

Prior to the main analysis, the content validity of the form was reviewed by an expert in mathematics education to ensure it covered all necessary dimensions of the research questions.

2.2. Data Analysis

The method used to analyze the data collected from the studies within the scope of the research is descriptive content analysis. Descriptive content analysis is a method used to obtain general results on a subject by utilizing descriptive statistics and identifying findings related to the frequencies and percentages of the data obtained (Dinçer, 2018).

Data used in the descriptive content analysis method can be examined from both quantitative and qualitative perspectives (Dinçer, 2018). In line with the purpose of this research,

the data were analyzed using a systematic qualitative approach. Following the four-step framework proposed by Yıldırım and Şimşek (2011), the analysis was conducted as follows:

Coding of Data: The full texts of the studies were examined, focusing specifically on the 'Method', 'Abstract', and 'Conclusion' sections. Structural characteristics were coded deductively based on the research questions. For instance, regarding research methods, studies were coded under categories such as 'Quantitative', 'Qualitative', or 'Mixed'. Similarly, for sample groups, codes such as 'Primary School', 'Middle School', 'High School', and 'Undergraduate' were generated based on the educational level of the participants.

Identifying Themes: For the analysis of research aims and outcomes, an inductive approach was adopted. Codes were derived directly from the data. For example, if a study focused on instructional strategies, it was coded under the theme 'Teaching the Concept of Infinity'; if it aimed to reveal student errors, it was coded as 'Determination of Misconceptions'.

Organizing and Interpreting: All generated codes were organized into frequency tables using MS Excel to reveal trends and patterns in the literature."

Following the coding of the data from the 224 studies identified in line with the researcher's objective using the "Study Evaluation Form," percentage and frequency calculations were performed using MS Office Excel, and tables and graphs were created. Thus, an attempt was made to identify the trends in studies on the concept of infinity in mathematics education in the literature.

2.2.1.Validity and Reliability

To ensure the validity of the study, the coding categories were based on the relevant literature (Fischbein, 2001) and checked by an expert in mathematics education. For the reliability of the analysis, a second researcher coded approximately 20% of the data independently. The consistency between the two coders was calculated using the formula proposed by Miles and Huberman (1994): [Reliability = Agreement / (Agreement + Disagreement) x 100]. The inter-coder reliability was calculated as 92%. Any disagreements were resolved through discussion to reach a consensus.

FINDINGS

This section contains subheadings related to the studies included in the research. The headings used in the tables and figures in the subheadings are distributed according to the publication years of the studies, publication type, research method, sample type, sample size, places where the studies were published, ındexing status, data collection tools, and distribution according to the aims and results of the studies.

3.1. Findings related to the first sub-problem of the studies on infinity

The first sub-problem analysed was 'How is the distribution of the studies on the concept of infinity according to the years of publication?'.The frequency and percentage values of the distribution of the analyzed studies by publication year are presented in Table 1.

Table 1

Distribution of Studies on Infinity by Publication Year

Year of Publication	Frequance (f)	Percentage (%)	Sample Study Codes
1956-1960	3	1.34	K1, M1, M2
1961-1965	2	0.89	M3, M4
1966-1970	1	0.45	M5

1971-1975	3	1.34	M6, M7, K2
1976-1980	8	3.57	M8, M9, M10, B1, M11...
1981-1985	7	3.13	M15, M16, M17, M18, K3...
1986-1990	14	6.25	DR.1, B2, K4, K5, M21...
1991-1995	10	4.46	B5, M29, M30, M31, M32...
1996-2000	14	6.25	M37, DR.2, B7, M38, M39...
2001-2005	35	15.63	M48, M49, M50, M51...
2006-2010	41	18.30	B15, B16, B17, B18, B19...
2011-2015	33	14.73	M99, M100, M101, M102…
2016-2020	24	10.71	K11, M128, M129, M130...
2021-2023	29	12.95	M145, M147, M146, M148...
Total	224	100	

It was observed that the studies contributing most to the literature numbered 41 and accounted for 18.30% between 2006 and 2010. It was determined that the study contributing least to the literature was 1 in number and accounted for 0.45% between 1966 and 1970.

3.2. Findings related to the second sub-problem of the studies on infinity

The second sub-problem analysed was ‘How is the distribution of studies on the concept of infinity according to publication types?’.The frequency and percentage values of the distribution of the analyzed studies by publication type are presented in Table 2.

Table 2

Distribution of Studies on Infinity by Publication Type

Publication Type	Frequance (f)	Percentage (%)	Sample Study Codes
Article	169	75.45	M1, M2, M3, M4, M5...
Book	13	5.80	K1, K2, K3, K4, K5...
Conference paper	24	10.71	B1, B2, B3, B4, B5...
Doctoral dissertations	7	3.13	DR.1, DR.2, DR.3, DR.4...
Master's theses	11	4.91	YL.1, YL.2, YL.3, YL.4...
Total	224	100	

Of the analyzed studies, 169 (75.45%) were articles, 24 (10.71%) were conference papers, 13 (5.80%) were books, 11 (4.91%) were master's theses, and 7 (3.13%) were doctoral dissertations. It can be seen that articles constitute the majority of the studies.

3.3. Findings related to the third sub-problem of the studies on infinity

The third sub-problem analysed was ‘How is the distribution of the studies on the concept of infinity according to the research method used?’.The frequency and percentage values of the distribution of the analyzed studies, categorized by the research method used, are presented in Table 3.

Table 3

Distribution of Studies on Infinity According to the Research Method Used

Research Method	Frequency (f)	Percentage (%)	Sample Study Codes
Qualitative	190	84.82	K1, M1, M2, M3...
Quantitative	8	3.57	M15, M34, M66...
Quantitative and Qualitative (Mixed)	26	11.61	DR.1, M36, M41, M44...
Total	224	100	

Of the analyzed studies, 190 (84.82%) employed qualitative research approaches, making the most significant contribution to the field. In contrast, 26 (11.61%) employed mixed research methods, and 8 (3.57%) utilized quantitative research methods, making a smaller contribution to the field.

3.4. Findings related to the fourth sub-problem of the studies on infinity

The fourth sub-problem analysed was 'How is the distribution of the studies on the concept of infinity according to the type of sample used?'. The frequency and percentage values of the distribution of the analyzed studies, categorized by the type of sample used, are presented in Table 4.

Table 4

Distribution of Studies on Infinity According to the Type of Sample Used

Sample Type	Frequency (f)	Percentage (%)	Sample Study Codes
Document	87	31.87	K1, M1, M2, M3, M4...
Primary School	14	5.13	M48, B16, M96, M82...
Middle School	35	12.82	M17,M25,M41,DR.4...
High School	41	15.02	M2, M28, M59, B13...
University	72	26.37	M8, M9, B10, M94...
Postgraduate	12	4.40	M142, M166, YL.4, DR.5...
Teacher	11	4.03	M75, M140, YL.3, B21...
Unspecified	1	0.37	DR.2
Total	273	100	

Of the analyzed studies, 87 used documents (31.87%), 72 used university students (26.37%), 41 used high school students (15.02%), 35 used middle school students (12.82%), 14 used primary school students (5.13%), 12 used graduate students as a sample at 4.40%, and 11 used teachers as a sample at 4.03%. One study did not specify the type of sample.

3.5. Findings related to the fifth sub-problem of the studies on infinity

The fifth sub-problem analysed was 'How is the distribution of the studies on the concept of infinity according to the sample size used?'.The frequency and percentage values of the distribution of the analyzed studies, categorized by sample size, are presented in Table 5.

Table 5

Distribution of Studies on Infinity According to the Sample Size Used

Sample Size	Frequency (f)	Percentage (%)	Sample Study Codes
1-10	30	21.90	M12, DR.2, B7, M48, M53...
11-20	11	8.03	M25, M56, B19, M75, M81...
21-30	10	7.30	M47, B10, M91, YL.5...
31-40	6	4.38	B6, M43, B20, M83, M127...
41-50	5	3.65	B3, M31, M70, M76, B21
51-60	8	5.84	M39, B9, M63, M79, M86...
61-70	3	2.19	M38, M140, M146
71-80	6	4.38	M115, M134, YL.10, M151...
81-90	1	0.73	M112
91-100	5	3.65	M36, B12, M108, YL.8…
101 and above	32	23.36	M10, DR.1, M28, M30...
Unspecified	20	14.60	M2, M6, M9...
Total	137	100	

Of the analyzed studies, 101 or more individuals (23.36%) and 1-10 individuals (21.90%) were more frequently preferred as sample sizes, while studies with unspecified samples accounted for 14.60%. Eighty-seven studies with document-type samples were excluded from this comparison.

3.6. Findings related to the sixth sub-problem of the studies on infinity

The sixth sub-problem analysed was 'How is the distribution of the studies on the concept of infinity according to the data collection tools used?'. The frequency and percentage values of the distribution of the analyzed studies in terms of the data collection tools used are presented in Table 6.

Table 6

Distribution of Studies on Infinity According to the Data Collection Tools Used

Data Collection Tool	Frequency (f)	Percentage (%)	Sample Study Codes
Document Analysis	87	23.84	K1, M1, M2, M3, M4...
Focus Group Interview	14	3.84	B9, M2, B14, B20, M78...
Observation	25	6.85	M25, M34, DR.2, B7, B10...
Survey	37	10.14	M8, DR.1, M36, M39, M41...
Interview	87	23.84	M9, M10, M15, M17, M30...
Open-ended Question	101	27.67	M12, M17, M28, M44...
Video Recording	7	1.92	DR.1, DR.2, M80, M101...
Voice Recording	6	1.64	M48, M56, M65, M94...
Unspecified	1	0.27	M16
Total	365	100	

Of the studies analyzed, 101 (27.67%) used open-ended questions, 87 (23.84%) used document analysis, 87 used interviews (23.84%), 37 used surveys (10.14%), 25 used observation (6.85%), 14 used focus group discussions (3.84%), 7 used video recording (1.92%), and 6 used audio recording (1.64%) as data collection tools. One study did not specify the data collection tool.

The publication profile of the studies was examined in two dimensions: their indexing status and the journals in which they were published. The first was the distribution of studies published in SSCI-indexed journals and other publications, and the second was the distribution of SSCI-indexed journals.

3.7. Findings related to the seventh sub-problem of the studies on infinity

The seventh sub-problem analysed was 'How are the examined studies distributed according to their indexing status and publication journals?'.The frequency and percentage values of the distribution of the analyzed studies, categorized by their publication in SSCI-indexed journals and other publications, are presented in Table 7.

Table 7

Distribution of Studies on Infinity According to Their Publication in SSCI-Indexed Journals and Other Journals

Indexing Status	Frequency (f)	Percentage (%)	Sample Study Codes
Journals with SSCI	60	26.79	M29, M33, M10, M13, M15…
Other Publications	164	73.21	DR.1, DR.2, YL.8,K1…
Total	224	100	

Of the analyzed studies, 60 (26.79%) were published in SSCI-indexed journals, and 164 (73.21%) were published in other publications.

The frequency and percentage values of the distribution of the analyzed studies, categorized by SSCI-indexed journals, are presented in Table 8.

Table 8

Distribution of Studies on Infinity According To Their Publication İn SSCI-İndexed Journals

Journal Name	Frequency (f)	Percentage (%)	Sample Study Codes
School Science and Mathematics	2	3.33	M3, M32
Educational Studies in Mathematics	33	55.00	M10, M13, M15, M17, M19...
Philosophy of Science	1	1.67	M24
British Journal for the Philosophy of Science	2	3.33	M29, M33
European Journal of Psychology of Education	1	1.67	M38
Journal for Research in Mathematics Education	7	11.67	M39, M43, M47, M53...
Historia Mathematica	1	1.67	M60
Science & Education	1	1.67	M67
Cognitive Psychology	1	1.67	M70
International Journal of Science and Mathematics Education	3	5.00	M77, M92, M134
Cognition and Instruction	3	5.00	M93, M97, M96
Synthese	3	5.00	M122, M153, M154
ZDM – Mathematics Education	2	3.33	M157, M166
Total	60	100	

When examining the distribution of the analyzed studies based on their publication in SSCI-indexed journals, the journal name “Educational Studies in Mathematics” has the highest number of studies with 33 articles, accounting for 55.00%.

3.8. Findings related to the eighth sub-problem of the studies on infinity

The eighth sub-problem analysed was ‘How are the primary research aims of the examined studies distributed in terms of their thematic focus?’. The frequency and percentage values of the distribution of the analyzed studies, categorized by their objectives, are presented in Table 9.

Table 9

Distribution of Studies on Infinity According to Their Objectives

Studies According to Their Purposes	Frequency (f)	Percentage (%)	Sample Study Codes
Determining perceptions of infinity	75	32.89	M145, M168, B14, YL.8...
Identifying misconceptions about infinity	14	6.14	M151, B2, M39, B20, M109...
Using the concept of infinity as a tool	43	18.86	M167, M38, M92, B16...
Providing information about the concept of infinity	85	37.28	K1, K3, M4, M11, M3...

Teaching the concept of infinity	11	4.82	M1, M25, M85, DR.6, YL.11...
Toplam	228	100	

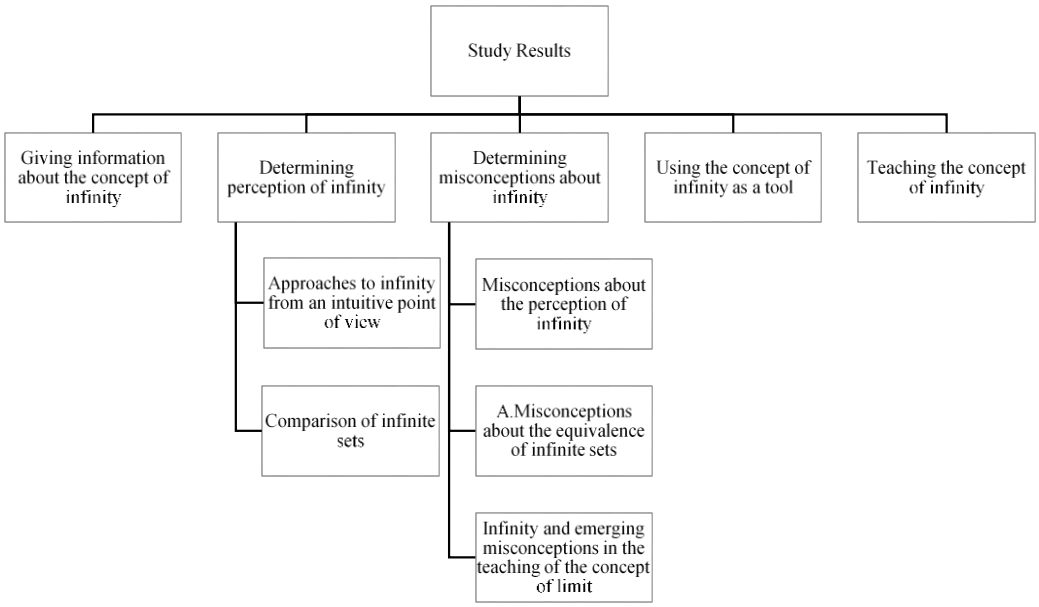
Of the analyzed studies, 85 (37.28%) provided information about the concept of infinity, 75 studies (32.89%) aimed to determine perceptions of infinity, 43 studies (18.86%) aimed to use the concept of infinity as a tool, 14 studies (6.14%) aimed to identify misconceptions related to infinity, and 11 studies (4.82%) aimed to present studies on teaching the concept of infinity.

3.9. Findings relating to the ninth sub-problem of studies on infinity.

The ninth sub-problem analysed was 'How are the key findings of the examined studies distributed across different thematic categories?'. We evaluated the studies included in our research under five headings according to their objectives: those providing information about the concept of infinity, those examining individuals' perceptions of infinity, those examining misconceptions related to the concept of infinity, those using the concept of infinity as a tool, and those associated with the teaching of the concept of infinity. We will discuss the results of the studies according to their objectives.

Figure 2

Results of Studies Are in Line With Their Objectives



3.9.1. Results achieved by studies written to provide information about the concept of infinity

Examining studies that aim to provide information about the concept of infinity reveals that the concept has been present in the human mind throughout history and that individuals have attempted to make sense of it in some way. The studies explain the questions raised by the concept in fields such as mathematics, philosophy, and logic since the Ancient Greek period. On the other hand, with the increase in knowledge gained over time, the perspective on the concepts of potential infinity and actual infinity, as mentioned by Aristotle, has evolved, and new concepts have emerged through the explanations of mathematicians in different periods. Like Galileo and Bolzano, Cantor also commented on the concept. Cantor, the founder of "Set Theory," addressed

infinity in a mathematical sense and put forward a new theory that combined infinity with finiteness and accepted the duality of infinity as potential and actual. Alongside this, the questions that arose brought paradoxes with them. Paradoxes that reveal infinity through events have been the subject of debate in the world of mathematics for many years. The concept of infinity, which also appears in topics such as limits, continuity, series, and infinite sums in analysis, has been influential in the interpretation of these topics. At this point, studies written for the purpose of providing information contain explanations about topics related to infinity.

3.9.2. Results of studies on determining the perception of infinity

Looking at the results of studies on determining the perception of infinity, the concept of infinity is a process that continues indefinitely (Bozkuş, 2014; İşleyen, 2013; Mamolo & Zazkis, 2008; Maria et al., 2009; Narlı & Narlı, 2012; Singer & Voica, 2003; Tall, 1981).

Another point examined was the effect of spoken language on mathematical discourse. It was noted that students who were more inclined to use the words "infinity" and "infinity sets" in their local language could influence their discourse on the concept of infinity (Kim et al., 2010). In addition, it has been stated that children's perceptions of the infinity of numbers in school should be considered not only in developmental terms (Fischbein, 1979; Monaghan, 2001) but also from a social construction perspective (Bartolini Bussi, 1989), and even didactic contracts and cultural environment (Boero et al., 2003). Another study found that individuals could not establish a relationship between language and mathematics; it was stated that they perceived the word "countable" as a new concept belonging to mathematics (Ünan & Doğan, 2011). On another point, it was found that the concepts of infinity, uncertainty, and indefiniteness were used interchangeably, and it was observed that prior knowledge influenced the use of symbolic language (Çelik & Aşkan, 2013). A different study (Jirotkova & Litter, 2003) showed that individuals did not use the adverbial and adjectival forms of the concept of infinity correctly, and in some cases, approached the idea that there is more than one infinity.

It has been noted that infinity is perceived as endless, uncertain, limitless, and a huge number (Aztekin, 2008; Bozkuş, 2014; Hewitt, 1995; Khasawneh, 2020; Tall, 2001) and that student statements such as "Infinity constantly increases, infinity constantly decreases" (Esposito, 1960; Yıldırım & Keşan, 2017). In addition, it has been found that there are problems in the perception and use of the infinity symbol, where individuals struggle to perceive infinity and therefore perceive it as impossible to perform any operation, or they view the infinity symbol as a number. A different study (Kolar & Cadez, 2012) showed that problems presented arithmetically are perceived as easier than those presented geometrically.

Another study (Aztekin, 2008) revealed that individuals hold ideas such as "infinite dimension," "infinite processes," and "infinite concept" regarding comprehension. It was concluded that the individuals participating in the study initially had a potential understanding of infinity and grasped the concept of actual infinity after receiving the given education. Similarly, Singer and Voica (2008) also stated that individuals' perceptions of infinity changed during the school process, shifting from an operational perception to a topological perception.

Another finding revealed that individuals' understanding of infinity is related to finite situations and that many students view infinite processes as a generalized form of finite processes (Barahmand, 2017). In addition, another study (Pehkonen et al., 2006) stated that individuals explained their ideas about the concepts of "infinite size" and "infinite multiplicity" in terms of actual infinity, and explained their ideas about the concept of "close to infinity" in terms of potential infinity. Examining the study's results, it was observed that individuals have intuitions about infinite sets, such as "repetitive process intuition/everything can be divided into two." However, it was stated that these intuitions, which are directed towards numerical sets, may deviate from the infinite nature of the relevant subdivision process when it comes to geometric

sets, due to their focus on the visual aspect. Therefore, the study shows that infinity is primarily understood as a potential (Tirosh, 1999).

The concepts of primary intuitions and secondary intuitions introduced to the literature by Fischbein (1987) can also be evaluated in the context of the perception of infinity. Schemas formed through everyday life experiences are primary intuitions, while those formed through new experiences after education are secondary intuitions. In addition to these two types of intuition, Tsamir (2001) introduced transient intuitions, which are a mixture of primary and secondary intuitions, to the literature. It has been stated that these types of intuitions are formed by external intervention. When examining infinity studies from an intuitive perspective, focusing specifically on natural numbers, thoughts such as "there is no largest natural number" have been encountered.

Another result frequently encountered in studies is that individuals possess an intuition of infinity before beginning formal education, and that this intuition can only be revealed through conversation. Studies conducted with younger age groups (Falk, 2010) have indicated that individuals in these age groups possess intuitions that enable them to discuss the concept of infinity. The results of the studies suggest that primary intuitions have a negative impact on formal learning (Barahmand, 2017; İşleyen, 2013; Khasawneh, 2020; Narlı & Narlı, 2012; Tall, 1991). Individuals of all ages, whether educated or not, have the concept of infinity in various areas. It has also been found that individuals' intuitions about the concept of infinity are influenced by their daily experiences, religious beliefs, and mathematical knowledge. When examining the subsequent approaches to infinity of individuals who have started their educational life, it has been observed that their theoretical understanding of the concept has changed, but their intuitive approaches have not (Aztekin, 2008). While some studies indicate that individuals' intuitions about infinity become regular around the age of 12 (Fishbein et al., 1979; Kolar & Čadež, 2012), other studies have concluded that, contrary to this, there is no consistent approach in this age group (Jirotka & Litter, 2004; Núñez, 1994). Studies have shown that age is not related to the intuition of infinity (Aztekin, 2008; Genç, 2021; Homaeinejad et al., 2021; Narlı & Narlı, 2012; Singer and Voica, 2003, 2007, 2008), and it has been stated that the connection with the concept of infinity should be part of education at a very early stage of concept learning. However, it has been emphasized that this connection should not be part of the curriculum (Singer & Voica, 2003, 2007, 2008).

In a study by Dubinsky et al. (2005), the APOS Theory was applied to propose a new explanation of how individuals think about the concept of infinity. When presented with an infinite process, the mental mechanisms of internalization and encapsulation enable individuals to form definitions about what happens after the process is completed. It was stated that the design of activities that include opportunities to think about repeated actions and require the transformation of infinite processes would be important in developing the student's perception of infinity and in creating an infinity schema. Other studies examining infinity from the perspective of APOS theory (Mamolo, 2009; Petty, 1996) have found that individuals perceive infinity as an inexhaustible process rather than an object structure. In a similar study, approaching infinity as a process has been interpreted as the perception of potential infinity in individuals (Bozkuş et al., 2015).

A review of the literature reveals studies that attempt to uncover the intuitive approach to the concept of infinity through various paradoxes. Focusing specifically on infinity paradoxes, the results of studies based on this foundation show that a survey conducted using Zeno's paradoxes (Núñez, 1994) determined that individuals under the age of 12 developed new intuitive perspectives on the concepts of infinitely small and infinitely large. It has been noted that when it comes to infinitely iterative processes, paradoxes affect students' reasoning about infinity (Apkarian et al., 2019; Ely, 2011). Another finding is that visualization becomes unreliable when used to discover the boundary or nature of an infinite process (Giaquinto, 1994). In a separate study (Wijeratne & Zazkis, 2016), the Green Alien and Thomson's Lamp paradoxes were

employed to investigate individuals' intuitive understanding of the concept of infinity. Here, the fact that the situation in the paradox is placed in an imaginary rather than a realistic context may be the reason why the supertask is seen as impossible (Wijeratne & Zazkis, 2016). Research on the formal aspects of infinity and paradoxes has also been found to be counterintuitive (Mamolo & Zazkis, 2008).

On the other hand, active learning on the concept of infinity and infinite sets, using different representations such as verbal (stories), graphical (drawings, number lines), and numerical (fraction sums), has been observed to facilitate individuals' understanding (Deihl & Markinson, 2019; Friedlander, 2010; Narlı & Başer, 2008).

Examining studies on the comparison of infinite sets reveals that the representation of infinite sets is effective in this context. It has been concluded that the one-to-one correspondence method facilitates the comparison of infinite sets. It has also been stated that set representations using geometric representations facilitate the transition to the one-to-one correspondence method (Tirosh & Tsamir, 1996; Tsamir, 2001).

Studies have found that individuals have approaches such as “Infinite sets have an equal number of elements,” “The whole is greater than the sum of its parts,” and “Infinity is an unquantifiable quantity” (Fischbein et al., 1981; Tirosh, 1991). Different studies have shown that individuals tend to count set elements, suggesting that the number of elements in finite sets can be determined, but this is not possible in infinite sets. Similar studies (Güven & Karataş, 2004; Narlı & Başer, 2008; Tirosh & Tsamir, 2006; Tsamir, 2001) have demonstrated that the properties of finite sets are also applicable to infinite sets. In contrast, another study (Na & Lee, 2006) confirmed that mathematically gifted individuals do not tend to apply the properties of finiteness when dealing with infinity.

In comparing infinite sets, ideas such as “infinite sets are incomparable,” “sets contain each other,” “the missing element must be determined,” and “infinity is unique, therefore all infinite sets are equal” have been observed (Güven & Karataş, 2004; Hewitt, 1995; Tirosh & Tsamir, 1996, 1999). A similar study indicated that thinking of infinite sets as single sets creates difficulties in the transition from finite to infinite (Monaghan, 1986; Tsamir & Dreyfus, 2002). Furthermore, it has been observed that some individuals holding these views use the concepts of countability, uncountability, and infinity interchangeably (Güven & Karataş, 2004). In a similar study, most students stated that countable finite sets and countable infinite sets exist, while also saying that uncountable finite sets and countable infinite sets do not exist (Göktepe Yıldız & Göktepe Körpeoğlu, 2018). Another study found that after receiving education and applying various techniques, individuals' misconceptions, such as “All infinite sets are equivalent,” were corrected and replaced with the idea that “No two infinite sets can be equivalent or compared” (Narlı & Başer, 2008).

3.9.3. Findings of studies identifying misconceptions about infinity

Studies aimed at determining perceptions of infinity naturally also reveal misconceptions about the concept of infinity. However, since some studies specifically seek to identify misconceptions, these are examined separately under a distinct heading. When examining studies conducted to identify misconceptions about the concept of infinity, it has been observed that the language used can be an obstacle to understanding the concept (Avitan, 2010; Tall, 1977; Ünan & Doğan, 2011). Situations that create difficulties in perceiving potential infinity are “egocentrism” and “lack of knowledge about the existence of infinite numbers.” In contrast, situations that cause difficulty in perceiving real infinity include “the big surpasses the small,” “excessive reinforcement of cardinal numbers,” and the perception that “the whole is greater than the sum of its parts” (Avitan, 2010).

Difficulties have been identified in defining and understanding the concept of infinity (Kanbolat, 2010). Additionally, the use of expressions such as “uncertain” and “undefined” for the concept of infinity can confuse (Oflaz & Polat, 2022). At this point, it has been noted that past experiences may contribute to the perception of conflict in the concept (Tall, 1977). Furthermore, it has been stated that elementary school teachers are influential in concept formation; however, they often lack sufficient knowledge (Martin & Wheeler, 1987). In another study (Fischbein & Schnarch, 1997), it was emphasized that misconceptions become stronger with age, while some weaken.

It has been stated that the difficulties students encounter in understanding the mathematical concept of infinity stem not only from epistemological barriers but also from didactic barriers related to the mathematical concept itself (Sbaraglı, 2006). Another study (Kanbolat, 2010) suggested that defining and understanding the concept of infinity, the expression “finite infinity,” the topic of different infinities, and the inability to define infinity entirely are epistemological obstacles. Furthermore, while it is evident that the result of the four operations performed with infinity will be infinite (Kanbolat, 2010), another study (Oflaz & Polat, 2022) suggests that the result will be indefinite.

Looking at the misconceptions regarding the equivalence of infinite sets: “All infinite sets have the same number of elements,” “Infinite sets cannot be equivalent,” “Only infinite sets that appear to have the same number of elements can be equivalent,” “No infinite set can be equivalent to its own subset.” (Narlı & Narlı, 2012), “incorrectly relating the concept of dimension to the concept of subset,” “treating an infinite set as finite,” “thinking that all infinities are the same,” “incorrectly defining infinite sets,” “treating the equivalence relation intuitively,” “Using the symbols $-\infty$ and ∞ as the smallest and largest numbers, respectively,” “Considering any correspondence between sets as sufficient for equivalence,” “Believing that a finite number of elements affects the equivalence of infinite sets,” “treating uncountable infinite sets as finite or countable sets,” “comparing infinite intervals based on their metric lengths” (Pala & Narlı, 2018).

In addition, studies show that individuals treat processes consisting of an infinite number of steps as if they will never end, and that this creates a difficulty corresponding to fundamental undergraduate mathematics concepts such as limits (Cottrill et al., 1996; Sierpinska, 1987; Tall & Schwarzenberger, 1978; Tall & Vinner, 1981). Infinity and the misconceptions that arise in teaching the concept of limits In Sierpinska's (1987) study, the “disregard of the concept of infinity” and “infinity being only a theoretical scheme, i.e., seen as meaningless” are mentioned, while Kadıoğlu and Kamali (2009), in their studies on the concept of limits, found that teachers treat infinity as a symbol. It is also common for individuals to make inappropriate generalizations from finite sets to infinite sets (Fischbein et al., 1979; Tirosh & Tsamir, 1996; Tsamir, 1999) (Ely, 2011).

3.9.4. Results achieved by studies using the concept of infinity as a tool

When examining studies that use the concept of infinity as a tool, it is seen that infinity is approached through abstract mathematics, limit teaching, the concept of infinitely small numbers, ∞, 0, and 1, the concepts of “uncertainty” and “indefiniteness,” the determination of individuals' proof images, and the teaching of irrational numbers.

There is a need to summarize the parts of the studies related to the concept of infinity. Barak (2007) stated that there are errors and misconceptions regarding the understanding and use of the concept of infinity in limit teaching. It was concluded that the approach created with the concept of infinitely small numbers in calculating limits is more effective than the recommended approach for teaching the concept of limits in the High School Mathematics Course Curriculum (Altun, 2009). Aşık (2010) added that students had difficulty distinguishing between expressions such as infinity, 1, 0, 1/0, ∞/∞, and $1/\infty$. It has been determined that irrational numbers are defined as “numbers whose digits after the decimal point cannot be counted,” “an infinite set of numbers

whose decimal parts are not periodic," or "numbers whose digits after the decimal point continue irregularly" (Adıgüzel, 2013). Williams (2001) reported that the concept of real infinity is a cognitive barrier to learning the definition of limits. It has been observed that students use the same criteria to compare the sizes of infinite sets for infinite series as well (Barahmand, 2021). The results of the study show that the concept of infinity, which is relevant in various fields such as numbers and operations, geometry, analysis, and abstract mathematics, plays an essential role in the teaching of different subjects, particularly in the learning of its definition.

3.9.5. Results of studies on teaching the concept of infinity

Various studies and different methods applied in the field of teaching the concept of infinity have been encountered in the literature. Each has been evaluated based on the results achieved. In his article, Perel (1958) stated that children experience the concept of "infinity" from an early age, both in everyday language and in mathematical language, and recommended that all mathematics teachers avoid mentioning infinity as much as possible and never define it incorrectly. He advocated the use of the words "infinite" and "endless" in explanations. It was emphasized that before discussing such debatable topics with students, it is necessary to ensure that they are old enough.

In a study using game techniques (Cordeiro, 1988), it was noted that to help young children grasp complex concepts such as infinity, they should be encouraged to "play" with ideas, allowing them to go as far as they can in the realm of imagination and fantasy, and striving to embed terminology and ideas in a shared value, thereby making the concept of infinity a part of their daily experience. In another study (Sacristán, 1997), which utilized a computer micro-world (Logo program) to explore infinite processes, such as infinite sequences and series, it was noted that students could reconstruct and redesign the tools and connections between them through the micro-world's programming activities and express themselves effectively. A similar study (Sacristán & Noss, 2008) found that the act of constructing images created with code on a computer is the most critical factor enabling coordination between representations, allowing students to reconcile sometimes conflicting information about infinity. Another study (Narlı, 2005) examined the effect of active learning and traditional education on success in numerical equivalence (comparing infinite sets) and found that students in the experimental group were more successful, in line with the methods used.

In another study (Ergene, 2021), teacher candidates evaluated which topics in the curriculum addressed the concept of infinity. The concept of infinity was associated with 30 learning outcomes in the fifth, sixth, seventh, and eighth grades; the most associated learning outcomes were found to be related to number sets, patterns, and irrational numbers. It was also observed that teacher candidates lacked sufficient knowledge about the teaching process of infinity. In another study (Akyol, 2023), two mathematical books, "The Cat in the Land of Numbers" and "Rescuing the Queen," were analyzed in terms of the concepts they addressed. It was noted that the book "The Cat in the Land of Numbers" included situations involving infinity, such as Cantor's Set Theory and Hilbert's Hotel, and that these concepts were conveyed to individuals through stories and pictures.

DISCUSSION

The findings of this research provide a general perspective on how infinity has been studied in mathematics education over the years. The analysis shows that research in this area became more frequent between 2006 and 2010. This increase could be related to the development of cognitive frameworks such as APOS and the growing interest in advanced mathematical thinking during that period (Tall, 2001).

One of the prominent results of this study is the frequent use of qualitative methods. This trend is consistent with the views of Sierpinska (1987) and Fischbein (2001), who state that examining an abstract concept like infinity often requires qualitative tools. Instead of focusing only on statistical results, researchers generally prefer to use interviews and open-ended questions to understand students' cognitive structures and misconceptions. This shows that understanding the internal process of learning infinity is seen as a priority in the literature.

Another important finding is that studies are mostly conducted with university students. This situation can be explained by the fact that infinity is more explicitly included in advanced mathematics courses at the undergraduate level (Tall, 2001). However, the limited number of studies at the elementary and middle school levels is a point that needs attention. Research has shown that younger students have intuitive ideas about infinity even if they do not know the formal definitions (Singer & Voica, 2003; Falk, 2010). If these early intuitions are not well understood, it may become more difficult to address the misconceptions that students develop in later years. Recent studies (Ergene, 2021; Narlı & Narlı, 2012) also suggest that although infinity is not a standalone topic in early curricula, it is implicitly related to number sets and patterns. Therefore, more research at lower grade levels could help in understanding how students' initial ideas form.

Finally, the results show that most studies are descriptive and focus on identifying misconceptions rather than teaching the concept. While the literature clearly defines common errors, such as the idea that "a whole is greater than its parts," there are fewer studies on specific teaching interventions. This suggests that future research could focus more on using tools like paradoxes or visualization to help students understand the concept better (Sacristan & Noss, 2008). In this sense, moving from identifying misconceptions to developing teaching methods could be a useful direction for the field.

CONCLUSION AND RECOMMENDATIONS

This study concludes that infinity remains a central yet challenging concept in mathematics education. The analysis of 224 studies shows that although there is a theoretical foundation in the literature, certain trends continue to dominate the field. Most research is qualitative and focuses on university students, which suggests that infinity is still largely viewed as an "advanced" topic. However, the findings also show that students experience a significant cognitive barrier when transitioning from intuitive, process-oriented ideas (potential infinity) to formal, completed structures (actual infinity). This persistent conflict between intuition and formal definitions is one of the most important results of the research.

Based on these results, several recommendations can be made for future research and practice. First, there is a clear need to focus more on younger students at the elementary and middle school levels. Since early intuitions form the basis of later learning, longitudinal studies could be very useful in understanding how these mental structures evolve over time. Additionally, instead of only identifying misconceptions, future research should focus more on teaching experiments. Using tools such as paradoxes, the history of mathematics, and digital visualization software—like WebQuests or dynamic geometry tools—can help students overcome epistemological obstacles more effectively. Expanding the scope of research beyond just "identifying errors" to include geometry or probability would also contribute to a more integrated understanding of the concept.

These findings also point to the potential benefits of introducing the concept of infinity more deliberately within the existing mathematics curriculum, starting from the middle school years.Introducing "intuitive infinity" activities as early as middle school could help students bridge the gap between "limitless processes" and "completed sets" before they reach advanced

calculus. Finally, teacher training programs should be designed to increase awareness of the common difficulties students face. Providing teachers with pedagogical tools to manage these intuitive conflicts, rather than just formal definitions, would strengthen the teaching process and help prevent long-term misconceptions.

REFERENCES

Adıgüzel, N. (2013). İlköğretim matematik öğretmen adayları ve 8. sınıf öğrencilerinin irrasyonel sayılar ile ilgili bilgileri ve bu konudaki kavram yanılgıları [Preservice elementary mathematics teachers' and eighth grade students' knowledge and misconceptions about irrational numbers] (*Master's thesis, Necmettin Erbakan Üniversitesi*). *Yükseköğretim Kurulu Ulusal Tez Merkezi*.

Akyol, F. M. (2023). Sayılar diyarındaki kedi ve kraliçeyi kurtarmak: İki çocuk matematik kitabının içeriğinin karşılaştırmalı analizi [A comparative analysis of the content of two children's mathematics books: Saving the cat in the land of numbers and the queen] (*Master's thesis, İhsan Doğramacı Bilkent Üniversitesi*). *Yükseköğretim Kurulu Ulusal Tez Merkezi*.

Allen, G. D. (2000). *The history of infinity*. Texas A&M University.

Apkarian, N., Tabach, M., & Dreyfus, T. (2019). The Sierpinski smoothie: Blending area and perimeter. *Educational Studies in Mathematics, 101*, 19–34. https://doi.org/10.1007/s10649-019-09889-4

Aşık, S. (2010). Tanımsızlık ve belirsizlik kavramlarının öğretmen ve öğretmen adaylarının görüş ve performansları bağlamında incelenmesi: 0, 1 ve ∞ ile yapılan işlemler [Investigation of undefined and indeterminate concepts in the context of teachers' and preservice teachers' views and performances: Operations with 0, 1, and ∞] (*Master's thesis, Marmara Üniversitesi*). *Yükseköğretim Kurulu Ulusal Tez Merkezi*.

Avitan, S. (2010). *Development of intuitive perceptions of children about the concept of infinity* [Çocukların sonsuzluk kavramına ilişkin sezgisel algılarının geliştirilmesi] [Doctoral dissertation, Haifa University]. ProQuest Dissertations & Theses Global.

Aztekin, S. (2008). Farklı yaş gruplarındaki öğrencilerde yapılanmış sonsuzluk kavramlarının araştırılması [Investigation of structured concepts of infinity in students of different age groups] (*Doctoral dissertation, Gazi Üniversitesi*). *Yükseköğretim Kurulu Ulusal Tez Merkezi*.

Barahmand, A. (2017). The boundary between finite and infinite states through the concept of limits of sequences. *International Journal of Science and Mathematics Education, 15*(3), 569–585. https://doi.org/10.1007/s10763-015-9697-3

Barahmand, A. (2021). Exploring students' consistency in their notions: the case of comparing infinite series. *International Journal of Mathematical Education in Science and Technology*, 52:7, 1049–1062, https://doi.org/10.1080/0020739X.2020.1736350

Barak, B. (2007). Limit konusundaki kavram yanılgılarının belirlenmesi [Determination of misconceptions about the concept of limit] *(Master's thesis, Gazi Üniversitesi). Yükseköğretim Kurulu Ulusal Tez Merkezi.*

Bartolini Bussi, M. (1989). Evaluation of teaching sequences that include individual and collective activities. *In Proceedings of the First Italian–German Bilateral Symposium on Didactics of Mathematics* (pp. 285–308). Pavia, Italy.

Boero, P., Douek, N., & Garuti, R. (2003). Children's conceptions of infinity of numbers in a fifth-grade classroom discussion context. *In Proceedings of PME* 27 (pp. 121–128). Honolulu, HI, USA.

Bozkuş, F. (2014). Ortaokul öğrencilerinin sonsuzluk kavrayışları [Middle school students' conceptions of infinity] *(Master's thesis, Dokuz Eylül Üniversitesi). Yükseköğretim Kurulu Ulusal Tez Merkezi.*

Bozkuş, F., Toluk Uçar, Z., & Çetin, İ. (2015). Ortaokul öğrencilerinin sonsuzluğu kavrayışları [Middle school students' conceptions of infinity]. *Turkish Journal of Computer and Mathematics Education (TURCOMAT), 6*(3), 506–531. https://doi.org/10.16949/turcomat.53890

Cohen, M., Manion, L., & Morrison, K. (2007). *Research methods in education* (6th ed.). Routledge.

Cordeiro, P. (1988). Playing with infinity in the sixth grade. *Language Arts, 65*(6), 557–566. http://www.jstor.org/stable/41411424

Cottrill, J., Dubinsky, E., Nichols, D., Schwinngendorf, K., Thomas, K., & Vidakovic, D. (1996). Understanding the limit concept: Beginning with a coordinated process schema. *Journal of Mathematical Behavior, 15,* 17–192.

Çalık, M., & Sözbilir, M. (2014). İçerik analizinin parametreleri [Parameters of content analysis]. *Education and Science, 39*(174), 33–38.

Çalık, M., Ünal, S., Coştu, B., & Karataş, F. Ö. (2008). Trends in Turkish science education. *Essays in Education,* Special Issue, 23–45.

Çelik, D., & Akşan, E. (2013). Matematik öğretmeni adaylarının sonsuzluk, belirsizlik ve tanımsızlık kavramlarına ilişkin anlamaları [Prospective mathematics teachers' understandings of the concepts of infinity, uncertainty, and undefined]. *Necatibey Faculty of Education Electronic Journal of Science and Mathematics Education, 7*(1), 166–190.

Deihl, S., & Markinson, M. P. (2019). Connecting the tangent function to cardinality: A method for introducing set theory to high school students. *Journal of Mathematics Education at Teachers College, 10*(2), 1–7. https://doi.org/10.7916/jmetc.v10i2.4190

Dinçer, S. (2018). Eğitim bilimleri araştırmalarında içerik analizi: Meta-analiz, meta-sentez, betimsel içerik analizi [Content analysis in educational science research: Meta-analysis, meta-synthesis, descriptive content analysis]. *Bartın University Journal of Faculty of Education, 7*(1), 176–190. https://doi.org/10.17556/erj.339616

Dubinsky, E., Weller, K., McDonald, M. A., & Brown, A. (2005). Some historical issues and paradoxes regarding the concept of infinity: An APOS-based analysis: Part 1. *Educational Studies in Mathematics, 58*(3), 335–359.

Dubinsky, E., Weller, K., McDonald, M. A., & Brown, A. (2005). Some historical issues and paradoxes regarding the concept of infinity: An APOS-based analysis: Part 2. *Educational Studies in Mathematics, 60,* 253–266.

Ely, R. (2011). Envisioning the infinite by projecting finite properties. *Journal of Mathematical Behavior, 30,* 1–18. https://doi.org/10.1016/j.jmathb.2010.12.001

Ergene, Ö. (2021). Öğretmen adayları gözünden sonsuzluk kavramı ve matematik dersi öğretim programı [The concept of infinity and mathematics curriculum from the perspective of preservice teachers]. *Van Yüzüncü Yıl University Journal of Education Faculty, 18*(2), 123–151. https://doi.org/10.33711/yyuefd.998263

Esposito, J. N. (1960). Infinite, finite, infinity. *The Mathematics Teacher, 53*(5), 395–397. http://www.jstor.org/stable/27956192

Falk, R. (2010). The infinite challenge: Levels of conceiving the endlessness of numbers. *Cognition and Instruction, 28*(1), 1–38. http://www.jstor.org/stable/27806340

Fischbein, E. (1987). *Intuition in science and mathematics: An educational approach*. Springer.

Fischbein, E. (2001). Tacit models and infinity. *Educational Studies in Mathematics, 48*(2–3), 309–329.

Fischbein, E., & Schnarch, D. (1997). The evolution with age of probabilistic, intuitively based misconceptions. *Journal for Research in Mathematics Education, 28*(1), 96–105. https://doi.org/10.2307/749665

Fischbein, E., Tirosh, D., & Hess, P. (1979). The intuition of infinity. *Educational Studies in Mathematics, 10*(1), 30–40.

Fischbein, E., Tirosh, D., & Melamed, U. (1981). Is it possible to measure the intuitive acceptance of a mathematical statement? *Educational Studies in Mathematics, 12*(4), 491–512.

Friedlander, A. (2009). Stories about never-ending sums. *Mathematics Teaching in the Middle School, 15*(5), 274–280. https://doi.org/10.5951/MTMS.15.5.0274

Genç, F. (2021). Ortaöğretim öğretmenlerinin sonsuzluk algıları [Secondary school teachers' perceptions of infinity] (*Unpublished master's thesis, Balıkesir Üniversitesi*).

Giaquinto, M. (1994). Epistemology of visual thinking in elementary real analysis. *British Journal for the Philosophy of Science, 45*(3), 789–813. https://doi.org/10.1093/bjps/45.3.789

Gökçe, O. (2006). İçerik analizi kuramsal ve pratik bilgiler [Content analysis: Theoretical and practical information] (1st ed.). *Siyasal Kitabevi*.

Göktepe Yıldız, S., & Göktepe Körpeoğlu, S. (2018). Exploring pre-service mathematics teachers' understandings of countability and infinity in WebQuest based learning environment. *European Journal of Education Studies, 5*(1), 94–120. https://doi.org/10.5281/zenodo.1403724

Güven, B., & Karataş, İ. (2004). Sonsuz kümelerin karşılaştırılması: Öğrencilerin kullandıkları yöntemler [Comparison of infinite sets: Methods used by students]. *Dokuz Eylül University Journal of Buca Faculty of Education, 15*, 65–73.

Hewitt, D., & A 2-year Mathematics PGCE Group. (1995). Infinity and arithmetic progressions. *Mathematics in School, 24*(3), 34–35. http://www.jstor.org/stable/30228207

Homaeinejad, M., Barahmand, A., & Seif, A. (2021). The relationship between notions of infinity and strategies used to compare enumerable infinite sets. *International Journal of Mathematical Education in Science and Technology, 53*(12), 3307–3325. https://doi.org/10.1080/0020739X.2021.1941362

İşleyen, T. (2013). Ortaöğretim öğrencilerinin sonsuzluk algıları [Secondary school students' perceptions of infinity]. *Kastamonu University Kastamonu Education Journal, 21*(3), 1235–1252.

Jayarajah, K., Saat, R. M., & Rauf, R. A. A. (2014). A review of science, technology, engineering & mathematics (STEM) education research from 1999–2013: A Malaysian perspective.

Eurasia Journal of Mathematics, Science and Technology Education, 10(3), 155–163. https://doi.org/10.12973/eurasia.2014.1072a

Jirotkova, D., & Littler, G. (2004). Insight into pupils' understanding of infinity in a geometrical context. In *Proceedings of the 28th Conference of the International Group for the Psychology of Mathematics Education* (pp. xx–xx). Bergen, Norway.

Jirotkova, D., & Littler, G. (2003). Students' concept of infinity in the context of a simple geometrical construct. In *Proceedings of the International Group for the Psychology of Mathematics Education* (Vol. 3, pp. 126–132).

Kadıoğlu, E., & Kamali, M. (2009). *Genel matematik* (4th ed.). Kültür Eğitim Vakfı Yayınları.

Kanbolat, O. (2010). Bazı matematiksel kavramlarla ilgili epistemolojik engeller [Epistemological obstacles related to some mathematical concepts] (*Master's thesis, Atatürk Üniversitesi*). *Yükseköğretim Kurulu Ulusal Tez Merkezi*.

Khasawneh, A. A. (2020). Analysis of infinity conceptions in mathematics among graduate, undergraduate, and secondary Jordanian students. *Psychology and Education, 57*(2), 84–90.

Kim, D., Sfard, A., & Ferrini-Mundy, J. (2010). Students' colloquial and mathematical discourses on infinity and limit. In *Proceedings of the 34th Conference of the International Group for the Psychology of Mathematics Education* (Vol. 3, pp. 161–168). Belo Horizonte, Brazil.

Kolar, V. M., & Cadez, T. H. (2012). Analysis of factors influencing the understanding of the concept of infinity. *Educational Studies in Mathematics, 80*(3), 389–412.

Lin, T. C., Lin, T. J., & Tsai, C. C. (2014). Research trends in science education from 2008 to 2012: A systematic content analysis of publications in selected journals. *International Journal of Science Education, 36*(8), 1346–1372.

Lune, H., & Berg, B. L. (2017). *Qualitative research methods for the social sciences*. Pearson.

Mamolo, A. M. (2009). *Glimpses of infinity: Intuitions, paradoxes, and cognitive leaps* (*Doctoral dissertation, Simon Fraser University*).

Mamolo, A., & Zazkis, R. (2008). Paradoxes as a window to infinity. *Research in Mathematics Education, 10*(2), 167–182.

Maria, K., Thanasia, M., Katerina, K., Constantinos, C., & George, P. (2009). Teachers' perceptions about infinity: A process or an object? In *Proceedings of CERME 6* (pp. 1771–1780). Lyon, France.

Martin, W. G., & Wheeler, M. M. (1987). Infinity concepts among preservice elementary school teachers. In *Proceedings of the International Group for the Psychology of Mathematics Education* (pp. 362–368).

Miles, M. B., & Huberman, A. M. (1994). *Qualitative data analysis: An expanded sourcebook* (2nd ed.). Sage.

Ministry of National Education (MEB). (2018). *Matematik dersi öğretim programı (İlkokul ve ortaokul 1–8. sınıflar)*.

Monaghan, J. (1986). Adolescents' understanding of limits and infinity (*Doctoral dissertation, University of Warwick*).

Monaghan, J. D. (2001). Young people's ideas of infinity. *Educational Studies in Mathematics, 48*(2–3), 239–257.

Moreno, L. E., & Waldegg, G. (1991). The conceptual evolution of actual infinity. *Educational Studies in Mathematics, 22*(3), 211–231.

Na, G., & Lee, E. (2006). Mathematically gifted students' conception of infinity. In *Proceedings of the International Group for the Psychology of Mathematics Education* (p. 412).

Narlı, S. (2005). Geliştirilen başarı testi ile geleneksel ve aktif öğrenme yöntemlerinin sayısal denklik konusu öğretiminde başarıya etkisinin değerlendirilmesi [Evaluation of the effects of traditional and active learning methods on success in teaching numerical equivalence] (*Unpublished doctoral dissertation, Dokuz Eylül Üniversitesi*).

Narlı, S., & Başer, N. (2008). Cantor's set theory and teaching prospective teachers. *International Journal of Environmental and Science Education, 3*(2), 99–107.

Narlı, S., & Narlı, P. (2012). Sonsuz sayı kümeleri ışığında ilköğretim öğrencilerinin sonsuzluk algı ve yanılgılarının belirlenmesi [Determination of primary school students' perceptions and misconceptions of infinity in the light of infinite number sets]. *Dokuz Eylül University Journal of Buca Faculty of Education*, (33), 122–133.

National Council of Teachers of Mathematics (NCTM). (2000). *Principles and standards for school mathematics*.

Núñez, R. (1994). Cognitive development and infinity in the small: Paradoxes and consensus. In *Proceedings of the Cognitive Science Society*. Lawrence Erlbaum Associates.

Oflaz, G., & Polat, K. (2022). Sonsuzluk, tanımsızlık ve belirsizlik kavramlarına ilişkin epistemolojik engellerinin belirlenmesi [Determining the epistemological obstacles regarding the concepts of infinity, undefined and uncertainty]. *Cumhuriyet International Journal of Education, 11*(2), 301–320. https://doi.org/10.30703/cije.993425

Özmantar, F. (2010). Sonsuzluk kavramı: Tarihsel gelişimi, öğrenci zorlukları ve çözüm önerileri [The concept of infinity: Historical development, student difficulties and suggestions for solutions]. In M. F. Özmantar, E. Bingölbali, & H. Akkoç (Eds.), *Matematiksel kavram yanılgıları ve çözüm önerileri* (2nd ed., pp. 151–180). Pegem Akademi.

Pala, O. (2020). İspat imajının dinamiklerinin sonsuz kümelerin denkliği bağlamında incelenmesi [Investigation of the dynamics of proof image in the context of equivalence of infinite sets] (*Doctoral dissertation, Dokuz Eylül Üniversitesi*). *Yükseköğretim Kurulu Ulusal Tez Merkezi*.

Pala, O., & Narlı, S. (2018). Matematik öğretmeni adaylarının sonsuz kümelerin denkliği ile ilgili ispatlama yaklaşımları ve yaşadıkları güçlükler [Pre-service mathematics teachers' proving approaches and difficulties regarding the equivalence of infinite sets]. *Turkish Journal of Computer and Mathematics Education (TURCOMAT), 9*(3), 449–475. https://doi.org/10.16949/turcomat.337191

Pehkonen, E., Hannula, M. S., Maijala, H., & Soro, R. (2006). Infinity of numbers: How students understand it. In *Proceedings of the 30th Conference of the International Group for the Psychology of Mathematics Education* (Vol. 4, pp. 345–352). Prague, Czech Republic.

Petty, J. A. (1996). *The role of reflective abstraction in the conceptualization of infinity and infinite processes* (*Doctoral dissertation*). ProQuest Dissertations & Theses Global. https://www.proquest.com/dissertations-theses/role-reflective-abstraction-conceptualization/docview/304262191

Sacristán, A. I. (1997). *Windows on the infinite: Creating meanings in a Logo-based microworld* (*Doctoral dissertation, University of London, Institute of Education*).

Sacristán, A. I., & Noss, R. (2008). Computational construction as a means to coordinate representations of infinity. *International Journal of Computers for Mathematical Learning, 13*, 47–70. https://doi.org/10.1007/s10758-008-9127-5

Sbaragli, S. (2006). Primary school teachers' beliefs and change of beliefs on mathematical infinity. *Mediterranean Journal for Research in Mathematics Education, 5*(2), 49–76.

Göktepe Yıldız, S., & Göktepe Körpeoğlu, S. (2018). Exploring pre-service mathematics teachers' understandings of countability and infinity in a WebQuest-based learning environment. *European Journal of Education Studies, 5*(1). https://doi.org/10.5281/zenodo.1403724

Sierpińska, A. (1987). Humanities students and epistemological obstacles related to limits. *Educational Studies in Mathematics, 18*(4), 371–397.

Singer, F. M., & Voica, C. (2008). Between perception and intuition: Learning about infinity. *Journal of Mathematical Behavior, 27*(3), 188–205.

Singer, F. M., & Voica, C. (2003). Perception of infinity: Does it really help in problem solving? In *Proceedings of the International Conference on Mathematics Education into the 21st Century Project*.

Singer, F. M., & Voica, C. (2007). Children's perceptions of infinity: Could they be structured? In *Proceedings of the Fifth Congress of the European Society for Research in Mathematics Education* (pp. 506–515).

Suri, H., & Clarke, D. (2009). Advancements in research synthesis methods: From a methodologically inclusive perspective. *Review of Educational Research, 79*(1), 395–430.

Tall, D. (1977). Conflicts and catastrophes in the learning of mathematics. *Mathematics Education for Teaching, 2*(4), 2–18.

Tall, D. (1980). Mathematical intuition, with special reference to limiting processes. In *Proceedings of the Fourth International Conference for the Psychology of Mathematics Education* (pp. 170–176). Berkeley, CA.

Tall, D. (1981). Intuitions of infinity. *Mathematics in School, 10*, 30–33.

Tall, D. O. (Ed.). (1991). *Advanced mathematical thinking*. Kluwer.

Tall, D. O., & Schwarzenberger, R. L. (1978). Conflicts in the learning of real numbers and limits. *Mathematics Teaching, 82*, 44–49.

Tall, D. O., & Vinner, S. (1981). Concept image and concept definition in mathematics with particular reference to limits and continuity. *Educational Studies in Mathematics, 12*(2), 151–169.

Tall, D. O., & Tirosh, D. (2001). Infinity—The never-ending struggle. *Educational Studies in Mathematics, 48*(2), 129–136.

Tall, D. O. (2001). Natural and formal infinities. *Educational Studies in Mathematics, 48*(2–3), 199–238.

Tirosh, D. (1991). The role of students' intuitions of infinity in teaching the Cantorian theory. In D. Tall (Ed.), *Advanced mathematical thinking* (pp. 201–214). Kluwer.

Tirosh, D. (1999). Finite and infinite sets: Definitions and intuitions. *International Journal of Mathematical Education in Science and Technology, 30*(3), 341–349. https://doi.org/10.1080/002073999287879

Tirosh, D., & Tsamir, P. (2006). Conceptual change in mathematics learning: The case of infinite sets. In *Proceedings of the 30th Conference of the International Group for the Psychology of Mathematics Education* (Vol. 1, pp. 159–161). Prague, Czech Republic.

Tirosh, D., & Tsamir, P. (1996). The role of representations in students' intuitive thinking about infinity. *International Journal of Mathematical Education in Science and Technology, 27*(1), 33–40. https://doi.org/10.1080/0020739960270105

Tsamir, P. (1999). The transition from comparing finite sets to comparing infinite sets: Teaching prospective teachers. *Educational Studies in Mathematics, 38*, 209–234.

Tsamir, P. (2001). When "the same" is not perceived as such: The case of infinite sets. *Educational Studies in Mathematics, 48*(2–3), 289–307.

Tsamir, P., & Dreyfus, T. (2002). Comparing infinite sets—A process of abstraction: The case of Ben. *Journal of Mathematical Behavior, 21*(1), 1–23.

Tsamir, P., & Tirosh, D. (1999). Consistency and representations: The case of actual infinity. *Journal for Research in Mathematics Education, 30*(2), 213–219. https://doi.org/10.2307/749611

Ültay, E., Akyurt, H., & Ültay, N. (2021). Sosyal bilimlerde betimsel içerik analizi [Descriptive content analysis in social sciences]. *IBAD Journal of Social Sciences,* (10), 188–201. https://doi.org/10.21733/ibad.863116

Ünan, Z., & Doğan, M. (2011). Sonlu ve sayılabilir sonsuz kümeler ve sayılamayan sonsuz kümelerin bir modellemesi [Finite and countable infinite sets and a modeling of uncountable infinite sets]. *Education Sciences, 6*(2), 1938–1950.

Wijeratne, C., & Zazkis, R. (2016). Exploring conceptions of infinity via super-tasks: A case of Thomson's lamp and Green Alien. *Journal of Mathematical Behavior, 41*, 1–17. https://doi.org/10.1016/j.jmathb.2015.11.002

Williams, S. R. (2001). Predications of the limit concept: An application of repertory grids. *Journal for Research in Mathematics Education, 32*(4), 341–367. https://doi.org/10.2307/749699

Yıldırım, A., & Şimşek, H. (2006). *Sosyal bilimlerde nitel araştırma yöntemleri* [Qualitative research methods in the social sciences] (5th ed.). Seçkin Yayıncılık.

Yıldırım, A., & Şimşek, H. (2011). *Sosyal bilimlerde nitel araştırma yöntemleri* [Qualitative research methods in the social sciences] (8th ed.). Seçkin Yayıncılık.

Yıldırım, K., & Keşan, C. (2017). Ortaokul öğrencilerinin üç nokta ve sonsuzluk kavramına ilişkin görüşleri [Middle school students' views on the ellipsis and the concept of infinity]. *Black Sea Journal of Social Sciences, 9*(16), 47–69.

Appendix 1. List of Sample Studies Examined in the Research

Study Code	Author(s)	Year	Title	Publication Type
K1	Barbel & Jean Piaget Inhelder	1958	The Growth Of Logical Thinking From Childhood To Adolescence...	Book
M1	W. M. Perel	1958	The Teaching Of Mathematical İnfinity	Article
M2	John N. Esposito	1960	Infinite, Finite, Infinity	Article

Study Code	Author(s)	Year	Title	Publication Type
M3	Margaret F. Willerding	1963	Infinity And Its Presentation At The High School Level	Article
M4	Guy Robinson & R. Harré	1964	Symposium: Infinity	Article
M5	Jaakko Hintikka	1966	Aristotelian Infinity	Article
M6	Evelyn Lawrence	1967	The Child's Conception Of Space By Jean Piaget...	Article
M7	Wıllıam P. Berlınghoff	1973	Numeratıon Bases And Infınıty	Article
M8	David Tall	1973	A Long-Term Learning Schema For Calculus And Analysis	Article
M9	David Tall	1976	Conflicts And Catastrophes İn The Learning Of Mathematics	Article
M10	D. O. Tall & R. L. E. Schwarzenberger	1978	Conflicts İn The Learning Of Real Numbersa And Limits	Article
M11	E. Fischbein, D. Tirosh & P. Hess	1979	The Intuition Of Infinity	Article
B1	David Tall	1980	Mathematical Intuition, With Special Reference To Limiting Processes	Conference Paper
M12	A. Gardiner	1980	One Hundred And One Ways To Infinity. Part 2	Article
M13	David Tall	1980	The Anatomy Of A Discovery İn Mathematics Research	Article
M14	David Tall	1980	The Notion Of Infinite Measuring Number...	Article
M15	Jonathan Lear	1980	Aristotelian Infinity	Article
M16	David Tall, Shlomo Vinner	1981	Concept Image And Concept Definition İn Mathematics...	Article
M17	David Tall	1981	Intuitions Of Infinity	Article
M18	E. Fischbein, D. Tirosh & U. Melamed	1981	Is It Possible To Measure The Intuitive Acceptance...?	Article
M19	Rouchier, A.	1983	Étude de la conceptualisation dans le système numéral	Article
B4	Waldegg, G.	1993	El infinito en la obra de Aristoteles	Conference Paper
DR.1	Monaghan, J.	1986	Adolescents' understanding of limits and infinity	PhD Dissertation
YL.5	Tsamir, P.	1999	The transition from finite to infinite sets...	Master Thesis

GENİŞLETİLMİŞ ÖZET

Giriş

Antik çağlardan bu yana çevresinde gördüğü nesneleri sayma, sınıflama gibi becerilere sahip insanoğlu, artma veya azalma gibi durumların bitmeden devam etmesi ihtimalini düşünmüş, nihayetinde sonsuzluk fikrini ortaya koymuştur (Allen, 2000). Çeşitli günlük yaşam durumları(tahta parçasının art arda bölünmesi, bir mesafenin her seferinde yarı kadarının gidilmesi vb.) sürekli olarak yenilendiğinde zihnen düşünülen durumlar fiziksel dünyada

anlamsız karşılıklar oluşturmuş; bu da sonsuzluk kavramının çelişkili yapısını ortaya koymuştur (Fischbein et al., 1979). Süreç içerisinde sonsuzluk kavramı "potansiyel sonsuzluk" ve "gerçek sonsuzluk" (Moreno & Waldegg, 1991), Kümeler Teorisi ile nesne olarak sonsuzluk (Tsamir & Dreyfus, 2002) gibi yaklaşımlarla ele alınmış; matematik camiasında çeşitli tartışmalara sebep olmuştur (Allen, 2000). Sonsuzluk kavramı ilkokuldan itibaren müfredatta tek başına bir konu olarak yer bulmasa da ileri seviye matematik konuları içerisinde bireylerin karşısına çıktığı; soyut yapısı nedeniyle kaçınılan bir kavram olduğu söylenebilir (Singer & Voica, 2003).Matematiksel süreç içerisinde pek çok noktada bulunmasına karşın, sonsuzluk kavramına yönelik ülkemizde yayınlanan az sayıda çalışmanın olması literatürde yer alan eğitimde sonsuzluk konulu çalışmaları geniş bir perspektif ile sunma ihtiyacını ortaya koymuştur. Bu çerçevede çalışmanın problemi "Eğitimde sonsuzluk kavramı ile ilgili yapılmış çalışmaların eğilimleri nasıldır?" şeklinde belirlenmiştir.

Yöntem

Bu çalışma ile, 1956-2024 yılları arasında eğitimde sonsuzluk konulu çalışmaların eğilimlerini belirlemek amaçlamaktadır. Araştırmada nitel araştırma yöntemlerinin etkileşimsiz desenlerinden biri olan betimsel içerik analizi yöntemi kullanılmıştır. Çalışmaya dahil edilen araştırmalar Ulusal Tez Merkezi, Google Scholer, Taylor ve Francis, Proquest Dissertations, Wiley Online Library, DergiPark, Researchgate.net, Academia, SSCI, ESCI, SCI, AHCI, SCIE, ERIC, Index Copernicus ve EBSCOhost isimli veri tabanlarında belirlenen 'sonsuzluk', 'sonsuzluk algısı', 'sonsuz kümeler', 'infinity', 'sense of infinity', 'infinite sets' anahtar kelimeleri kullanarak taranmıştır. Bu noktada alan yazın incelendiğinde Fischbein ve arkadaşları tarafından 1979 yılında ortaya koyulan çalışmada, bireylerin sonsuzluğu anlamlandırma çalışmalarının başlangıcı olarak Piaget ve Inhelder'in 1956 tarihli çalışmasının kabul edilebileceği dolaylı şekilde de olsa ileri sürülmüştür (Monaghan, 2001). Bundan yola çıkarak, araştırma kapsamında incelenecek olan çalışmalarda 1956 yılı referans noktası alınarak 2024 yılına kadar olan süreç taranmış; geniş bir perspektif sunulmaya çalışılmıştır. İncelemeye alınan çalışmaların belirlenmesinde; (1)1956-2024 yılları arasında olan, (2) Eğitimde sonsuzluk kavramı konulu, (3) Başlık ve özet kısmında Türkçe ve İngilizce olarak anahtar kelimeleri içeren ve (4) Açık erişimli olması kriterleri dikkate alınmıştır. Buna ek olarak (1) Açık erişimli olmayan, (2) Sonsuzluk kavramının yer aldığı süreklilik, sonsuz toplamlar gibi ileri matematik konularını içeren çalışmalar araştırmadan hariç tutulmuştur. Bu noktada yapılan tarama sonucunda, 1956-2024 yılları arasında eğitimde sonsuzluk konulu 224 çalışma araştırma kapsamında incelenmiştir. Araştırmacı amacı doğrultusunda belirlenen çalışmalar "Çalışma Değerlendirme Formu" kullanılarak kodlanmasının ardından MS Office Excel program kullanılarak yüzde ve frekans hesaplamaları yapılmıştır, tablo ve grafikler oluşturulmuştur. Form ile çalışmalar, yayın yılı, yayın türü, araştırma yöntemi, örneklem türü, örneklem büyüklüğü, çalışmaların yayın yapıldığı yer, çalışmaların tarandığı indeksler, veri toplama araçları, çalışmaların amaç ve sonuçlarına göre gruplandırılarak incelenmiştir.

Sonuç, Tartışma ve Öneriler

Araştırma kapsamasında elde edilen bulgulara bakıldığında, 1956-2024 yılları arasında incelenen çalışmaların yıllara göre dağılımında alan yazına en fazla çalışmanın 2006-2010 arası yıllarda olduğu görülmüştür. Çalışmaların yayın türlerine göre dağılımı değerlendirildiğinde ise makale türündeki çalışmaların ön plana çıktığı tespit edilmiştir. Çalışmalarda kullanılan araştırma yöntemine göre dağılımda nitel araştırma yöntemlerinin büyük çoğunluğu oluşturduğu söylenebilir. Bu durumun sonucu olarak görüşme, açık uçlu soru, doküman analizi gibi nitel veri toplama araçları daha çok kullanılmıştır. Örneklem türü doğrultusunda dağılıma bakıldığında katılımcı olarak üniversite öğrencilerinin en fazla olduğu görülmektedir. Bu noktada sonsuzluk kavramının limit, süreklilik, sonsuz kümelerin kıyaslaması gibi ileri seviye matematik konularının içerisinde karşımıza çıkması (Tall, 2001) çalışılan örneklem türündeki üniversite öğrencisi olması durumu ile uyuşmaktadır. Ancak araştırma kapsamında incelenen çalışmalar, erken yaş aralığında

bulunan bireylerin sezgisel olarak sonsuzluk hakkında bir ön bilgisinin bulunduğunu, buradan hareketle erken yaş grubundaki bireylerin de sonsuzluk algılarının incelenmesi gerektiğini belirtilmiştir (Falk, 2010). Diğer taraftan incelenen çalışma sonuçları, bireylerin formal eğitim öncesi sonsuzluk kavramına dair sahip oldukları zihinsel yapıların formal eğitim sürecini olumlu veya olumsuz yönde etkileyebileceğini göstermiştir (Tsamir, 2001). Çalışmalar örneklem büyüklüğü dağlımı açısından incelendiğinde 1-10 arası küçük gruplar ile 101 ve üzeri büyük gruplarda yapılan çalışmaların daha fazla olduğu görülmüştür. Tarandığı indeksler açısından bakıldığında, çalışmaların %26,79'unun SSCI kapsamındaki dergilerde, %73,21'inin ise diğer yayınlarda yer aldığı görülmektedir. Çalışmaların SSCI dergilerdeki yayınlarına göre dağılımı değerlendirildiğinde en çok yayının "Educational Studies in Matematics" isimli dergiye ait olduğu belirlenmiştir. Amaçlarına göre incelendiğinde en fazla bilgi vermek, devamında sonsuzluk algısını belirlemek ve sonsuzluk kavramını araç olarak kullanmak için çalışmaların yapıldığı tespit edilmiş; sonsuzluk kavramının öğretimi amacıyla ortaya koyulan çalışmaların en az orana sahip olduğu görülmüştür. Araştırma kapsamında ilgili literatüre bakıldığında sonsuzluk algısı, sonsuzluğa dair bireylerin sahip olduğu kavram yanılgıları, seriler, sonsuz toplar gibi ileri seviye konular içerisinde ele alış şekilleri, sonsuzluk kavramının farklı öğretim yöntemleri noktasında çalışmalara konu olduğu görülmüştür.

Elde edilen bulgular doğrultusunda sonsuzluk kavramına yönelik erken yaş grupları ile gerçekleştirilen çalışmaların daha az olduğu belirlenmiştir. Buradan hareketle ilkokul ve ortaokul gibi erken yaş grupları ile kavramın sezgisel boyutuna yönelik çalışmalar yapılabilir. Ayrıca makale yayın türüne göre daha kapsamlı çalışmalar olan tez çalışmaları ile alana katkı sağlanabilir. Matematik öğretim programları içerisinde kavrama ayrılan alanın dar olduğu belirtilebilir. Bu noktada öğretim programlarında sonsuzluk durumları içeren paradokslar yer verilerek kavramın çeşitli boyutlarla ele alınması sağlanabilir.